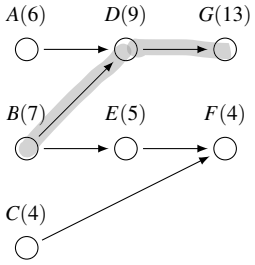
Name: Solutions score:_____ / 10

There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a non-programmable calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

1. Consider the following directed graph.



(a) (1 pt) What tasks need to be completed before task D can start?

A, B

(b) (1 pt) What tasks are available to be assigned at the beginning of the project?

A, B, C

(c) (1 pt) What is the critical path?

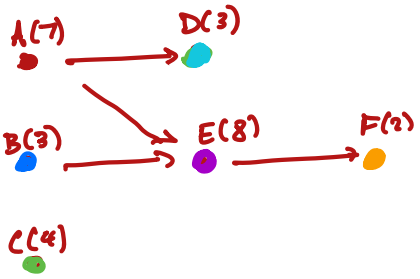
B → D → G

(d) (1 pt) What is the critical time? 29

2. Suppose you have a project consisting of the seven tasks described in the following table.

(a) (2 pts) Create a project digraph with vertices labeled. Draw it next to the table.

label/task	time	dependence
A	7	
B	3	
C	4	
D	3	A
E	8	A, B
F	2	E



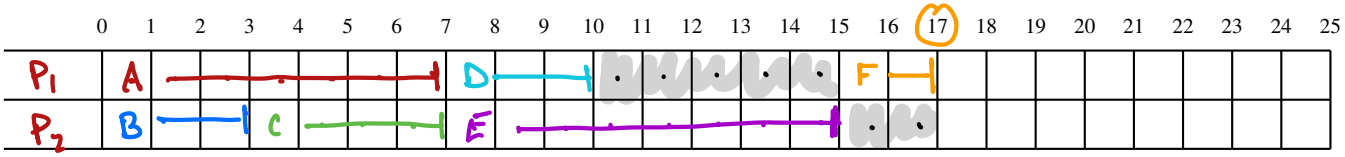
(b) (3 pts) Schedule the project with 2 processors using priority list

A B C D E F

Make sure you write clearly so that someone else can read your schedule!

The finishing time is 17 and the total idle time is 7.

time	0	3	7	10	15	17
ready	A B C	C	D E	—	F	—
done	—	B	A, C	D	E	F



(c) Create a priority list for this project based on the Decreasing-Time algorithm.

E	A	C	B	D	F
8	7	4	3	3	2